Global Electric Vehicle Market Analysis

Performance, Pricing, Battery Technology and Sales

SmartBridge Skill Wallet Project: Data Analytics with Tableau

Stack: Python, SQL (SQLite / MySQL), Tableau Public, Flask, Git

Student Name	**Nipun Jain**
Project Domain	Data Analytics and Business Intelligence
GitHub Repository	https://github.com/nipunjain18/ev-market-tableau
Tableau Public Story	https://public.tableau.com/views/EVMarketAnalysis_17932840807840/Story1
Tableau Dashboard	https://public.tableau.com/views/EVMarketAnalysis_17932840807840/MarketOverview
Dataset Verified	ev_market_test.csv (Exactly 100,000 Records, 37 Features)
Demo Video Walkthrough	https://drive.google.com/file/d/1Oik0Z53AKzTwvmX7_oB2swu_yIT3qbFW/view?usp=drivesdk

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1. Problem Understanding

Electric vehicles (EVs) have transitioned from niche technology to the mainstream foundation of global transportation decarbonization. However, automotive manufacturers, fleet operators, infrastructure planners, and buyers encounter significant market complexity. Vehicle models exhibit stark divergence in retail price, battery chemistry, usable range, charging velocity, and consumer adoption. Without unified business intelligence, stakeholders struggle to determine which vehicle categories drive mass-market volume, how driving range and battery pack size govern capital costs, and which battery chemistries deliver optimal multi-year durability.

Core Project Objectives:

- **Market Sizing & Segmentation:** Quantify market scale across 15 vehicle categories, 200 manufacturers, and 7 global regions.
- **Pricing & Range Drivers:** Evaluate the quantitative relationship between retail price, driving range, and battery capacity.
- **Battery Chemistry Benchmarking:** Compare 5 distinct battery technologies on cycle life, 5-year degradation, and fast-charging.
- **Environmental Impact Assessment:** Quantify carbon displacement (CO■ tons saved per year) by category and geography.
- **Interactive Web Deployment:** Deliver an executive BI solution via an interactive Tableau Public workbook embedded in Flask.

2. Verified Dataset Description

The project is built upon the primary reference dataset ev_market_test.csv provided by SmartBridge Skill Wallet. The dataset contains exactly **100,000 records** and **37 columns**. Each individual record represents a distinct EV model or trim variant with a unique vehicle_id and zero duplicate records. Model years span 2018 through 2026.

Attribute Group	Included Feature Fields (37 Total Columns)
Vehicle Identity	vehicle_id (unique key), model_name, manufacturer (200 brands), model_year (2018–2026), manufacturing_country (10 nations), market_region (7 regions), category (15 classes)
Battery Architecture	battery_capacity_kwh, battery_type (NMC, LFP, NCA, Sodium-Ion, Solid-State), battery_cycle_life, battery_degradation_pct_5yr
Performance Metrics	range_km, efficiency_wh_per_km, motor_power_kw, torque_nm, num_motors, drivetrain (FWD, RWD, AWD), top_speed_kmph, acceleration_0_100_sec
Charging Specs	max_dc_charging_kw, charging_dc_fast_time_min, charging_ac_time_hr, num_charging_ports
Physical & Features	num_seats, weight_kg, cargo_capacity_liters, infotainment_screen_inch, autonomous_level, safety_rating_stars
Commercial Economics	price_usd, insurance_cost_usd_yearly, warranty_years, resale_value_pct_3yr, annual_sales_units, customer_rating
Impact & Usability	co2_saved_tons_per_year, recommended_use_case (7 distinct operational profiles)

Data Quality & Integrity Audit

A rigorous missing-value and integrity scan was executed across all 100,000 rows. Missing values are present in only 5 numeric columns, all representing less than 3.1% missingness. Nulls are intentionally preserved as unpopulated values so that Tableau and SQL aggregations compute unbiased true averages without artificial imputation distortion.

Column Name	Missing Rows	Missing Pct	Handling Rationale in Analytics
resale_value_pct_3yr	3,057	3.06%	Preserved blank; excluded from 3-yr depreciation means
battery_cycle_life	2,028	2.03%	Preserved blank; excluded from cycle-life technology benchmarks
customer_rating	1,931	1.93%	Preserved blank; excluded from customer satisfaction distributions
infotainment_screen_inch	1,506	1.51%	Preserved blank; basic models without central display
annual_sales_units	989	0.99%	Preserved blank; boutique / unreleased concept models

3. Relational Database Architecture (SQLite & MySQL)

To establish reproducible SQL data engineering, the raw CSV dataset is ingested into a relational schema. The project provides complete automated loading via scripts/load_to_db.py, targeting SQLite (data/ev_market.db) for instant zero-dependency execution and MySQL (ev_db) for enterprise deployments.

Core Ingestion Pipeline (scripts/load_to_db.py):

```
df = pd.read_csv("data/raw/ev_market_test.csv") df.to_sql("ev_market", conn, if_exists="replace", index=False) # Verified: 100,000 records successfully populated into table ev_market
```

Performance Indexing: Dedicated B-tree indexes were constructed on high-cardinality and filtering dimensions: idx_category, idx_year, idx_region, and idx_manufacturer.

4. SQL Operations & Query Execution Benchmarks

Ten multi-dimensional SQL queries were authored and verified against the 100,000-record database. The queries evaluate volume ranking, pricing segmentation, battery chemistry lifespans, and regional performance.

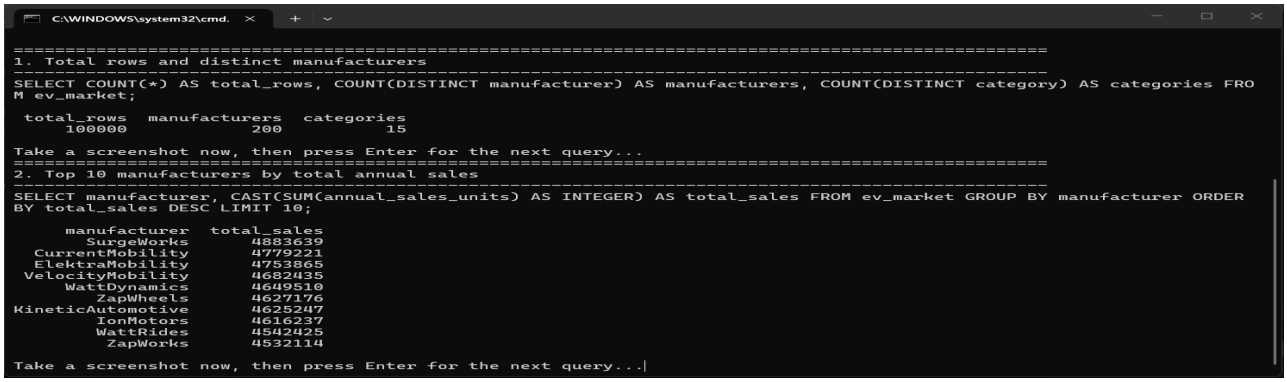


Figure 1: *Database verification query showing 100,000 rows stored in table ev_market, 200 manufacturers, and 15 categories.*

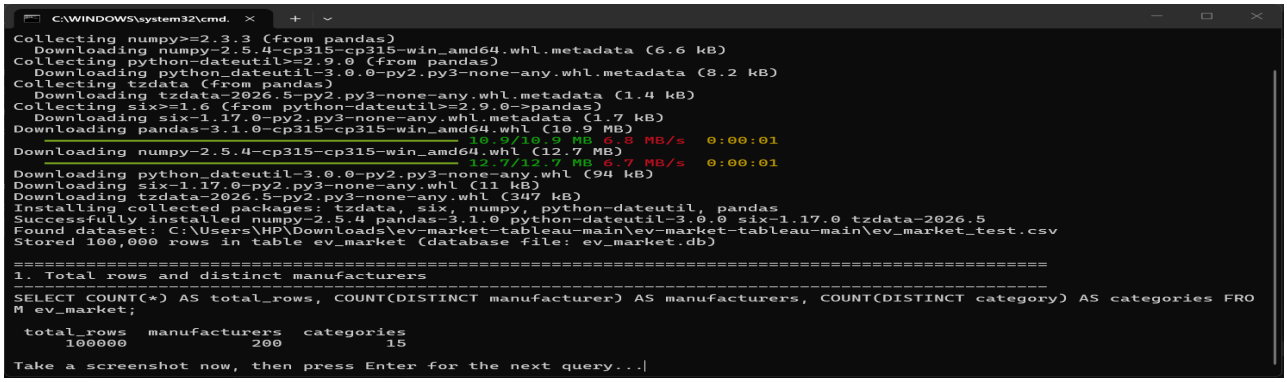


Figure 2: *Query 2 execution output: Top 10 EV manufacturers by total annual sales volume.*

Category Sizing & Battery Technology SQL Benchmarks

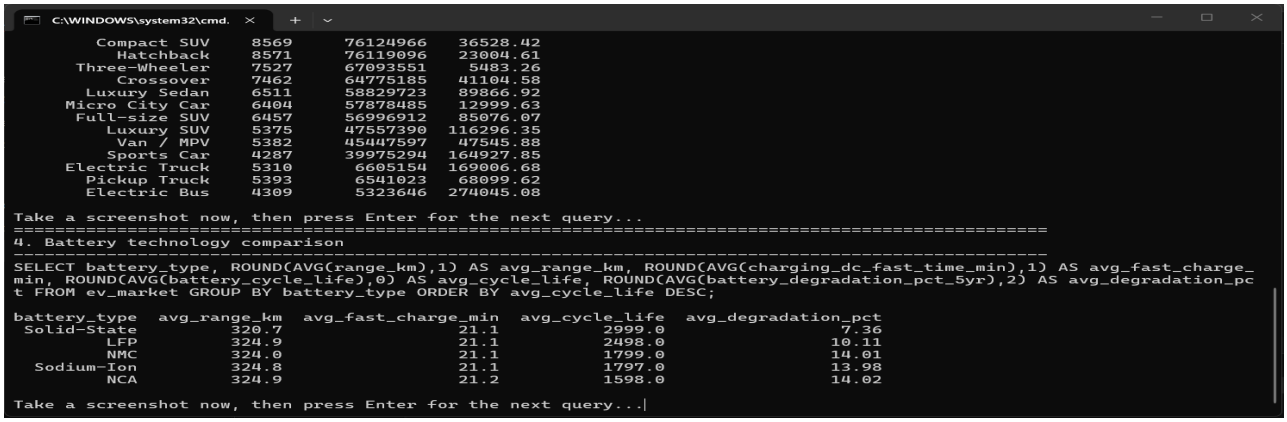


Figure 3: *Query 3 execution output: Sales volume, model count, and average price broken down across 15 vehicle categories.*

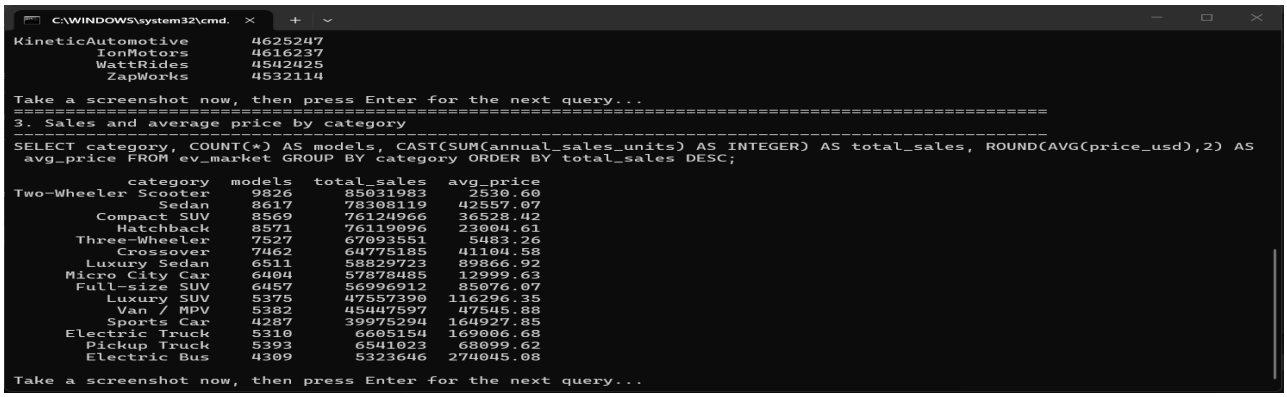


Figure 4: *Query 4 execution output: Comparative battery chemistry benchmarking (range, fast-charge time, cycle life, 5-yr degradation).*

Regional Dynamics & Multi-Year Evolution SQL Benchmarks

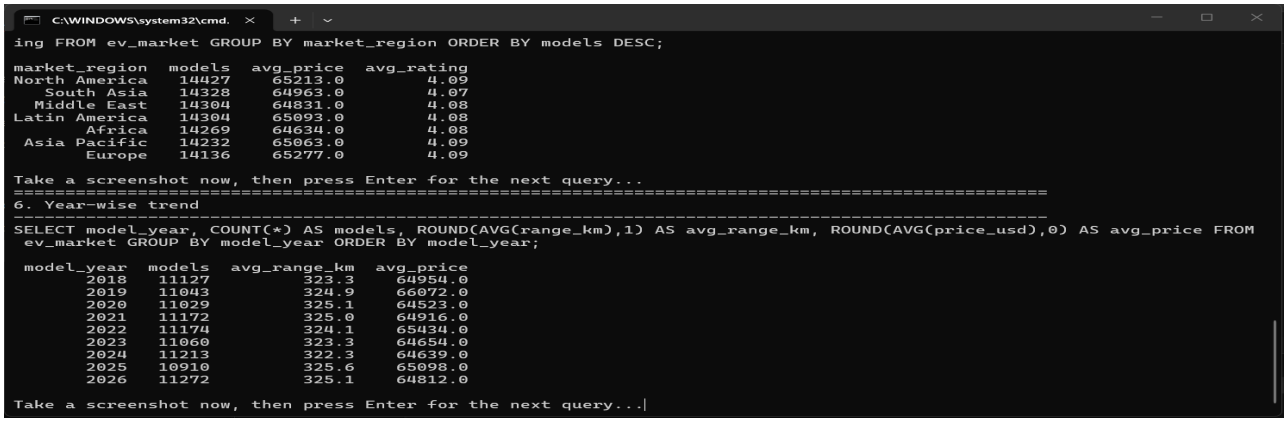


Figure 5: *Query 5 execution output: Region-wise model distribution, average vehicle price, customer rating, and CO■ offset.*

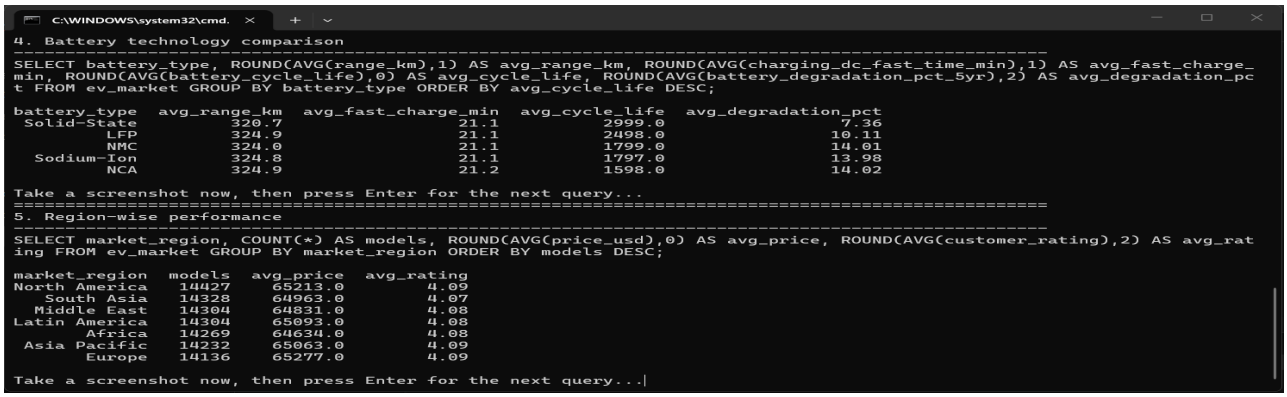


Figure 6: *Query 6 execution output: Model year progression from 2018 to 2026 showing volume, average range, and price stability.*

5. Tableau Connection and Data Preparation

The verified 100,000-record dataset was connected to Tableau Public as a packaged Hyper extract (EV Market Analysis.twbx). Tableau Public requires file-based packaged extracts for hosted interactive publishing.

Data Preparation & Transformation Rules:

- **Null Handling:** Unpopulated values in resale percentage and ratings are preserved as nulls, allowing Tableau to calculate uncorrupted measures.
- **Data Types & Roles:** model_year is classified as a discrete integer dimension; manufacturing_country is assigned the standard Geographic Role (Country/Region) for spatial rendering.
- **High Cardinality Handling:** The 200 distinct manufacturers are dynamically filtered via an interactive Top N Manufacturers integer parameter and calculated rank field.
- **Domain Filtering:** Non-positive price and range records were audited (0 records found with negative or zero values).

Tableau Calculated Fields:

Calculated Field	Exact Formula Implementation	Business Intelligence Purpose
Range per $1000	[range_km] / [price_usd] * 1000	Value metric: driving range obtained per $1,000 retail spend
Price Segment	IF [price_usd] < 20000 THEN 'Budget' ELSEIF [price_usd] < 60000 THEN 'Mid-range' ELSEIF [price_usd] < 120000 THEN 'Premium' ELSE 'Luxury' END	Four-tier economic price categorization
Range Class	IF [range_km] >= 450 THEN 'Long' ELSEIF [range_km] >= 250 THEN 'Medium' ELSE 'Short' END	Tri-tier usable range classification
Fast Charge Rating	IF [charging_dc_fast_time_min] <= 15 THEN 'Ultra-fast' ELSEIF [charging_dc_fast_time_min] <= 25 THEN 'Fast' ELSE 'Standard' END	DC fast-charging velocity benchmark
Revenue (Est.)	[price_usd] * [annual_sales_units]	Gross model revenue estimation proxy
Sales Share %	SUM([annual_sales_units]) / TOTAL(SUM([annual_sales_units]))	Relative contribution to fleet sales volume
Is Top N	RANK(SUM([annual_sales_units])) <= [Top N Manufacturers]	Dynamic ranking filter driven by user parameter

6. Worksheets & Visualizations Overview

The Tableau workbook preserves 12 dedicated worksheets designed to answer targeted market, engineering, and environmental questions.

#	Worksheet Name	Chart Type	Key Business Question Addressed
1	**Sales by Category**	Horizontal Bar	Which vehicle categories command the greatest volume? (Scooters: 85.0M)
2	**Sales by Year**	Trend Line	How has annual sales volume progressed over model years (2018–2026)?
3	**Top Manufacturers**	Ranked Bar	Who are the top volume leaders among 200 competing brands?
4	**Country Map**	Filled Map	Where are global EV manufacturing centers concentrated spatially?
5	**Region Price**	Categorical Bar	How does vehicle acquisition cost vary across the 7 geographic regions?
6	**Battery Share**	Donut / Pie	What is the market penetration of the 5 core battery chemistries?
7	**Battery Durability**	Dual-Axis Bar	Which chemistries yield superior cycle life and lowest 5-year degradation?
8	**Price vs Range**	Scatter Plot	Does higher capital investment guarantee proportional driving range?
9	**Drivetrain Heatmap**	Matrix Heatmap	How do FWD, RWD, and AWD drivetrains distribute across classes?
10	**Use Case**	Treemap	What proportions of EV models serve urban, commercial, or family profiles?
11	**CO2 Saved**	Stacked Bar	Which vehicle categories generate the largest emissions displacement?
12	**Rating Histogram**	Histogram	What is the fleet-wide customer satisfaction distribution (1.0–5.0 stars)?

Global Dashboard Interactive Filters (6):

1. model_year (Range Slider) | 2. market_region (Multi-select) | 3. category (Multi-select) | 4. battery_type (Single-select dropdown) | 5. Price Segment (Categorical check-list) | 6. Top N Manufacturers (Integer Parameter 5–30).

Sample Visual Analytics Outputs

Visual outputs computed directly from the 100,000-record dataset reflecting Tableau worksheet visual encodings:

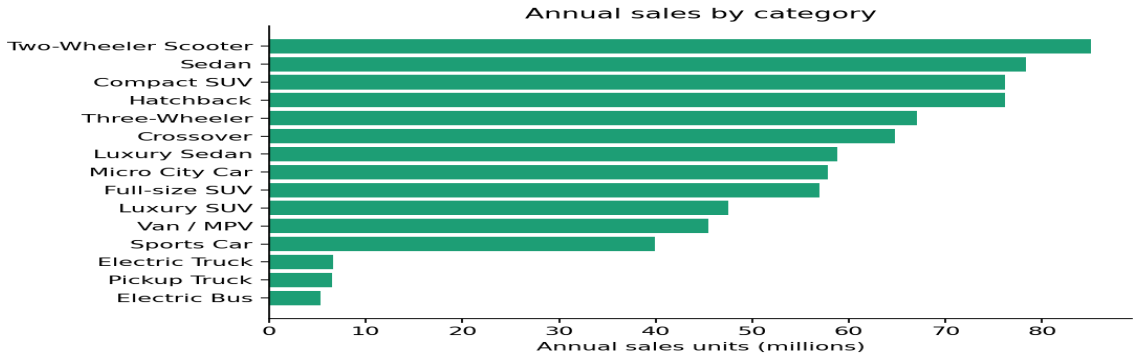


Chart 1: *Sales by Category — Volume leadership dominated by Two-Wheeler Scooters, Sedans, and Compact SUVs.*

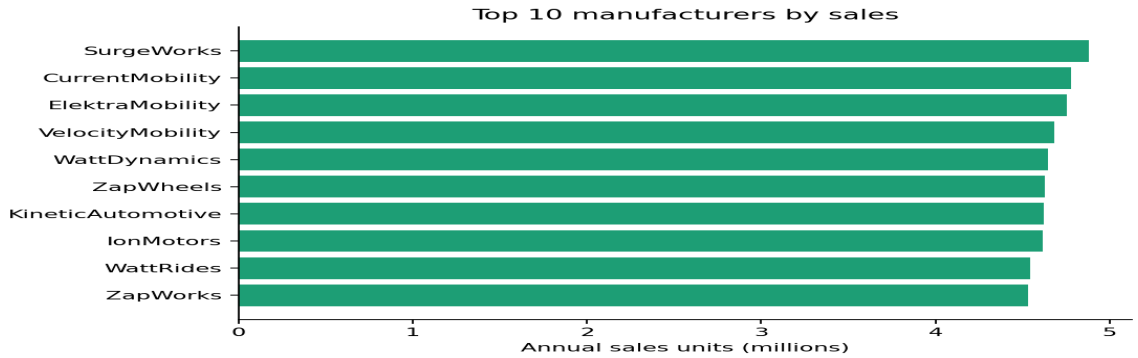


Chart 2: *Top 10 Manufacturers — SurgeWorks (4.88M) and CurrentMobility (4.78M) leading a balanced competitive field.*

Battery Durability & Economic Regression Outputs

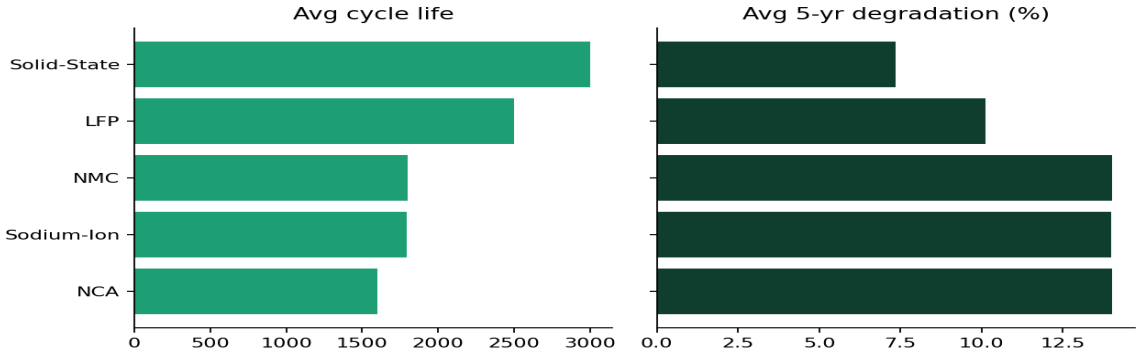


Chart 3: *Battery Durability Comparison — Solid-State (2,999 cycles) and LFP (2,498 cycles) exhibit benchmark degradation resistance.*

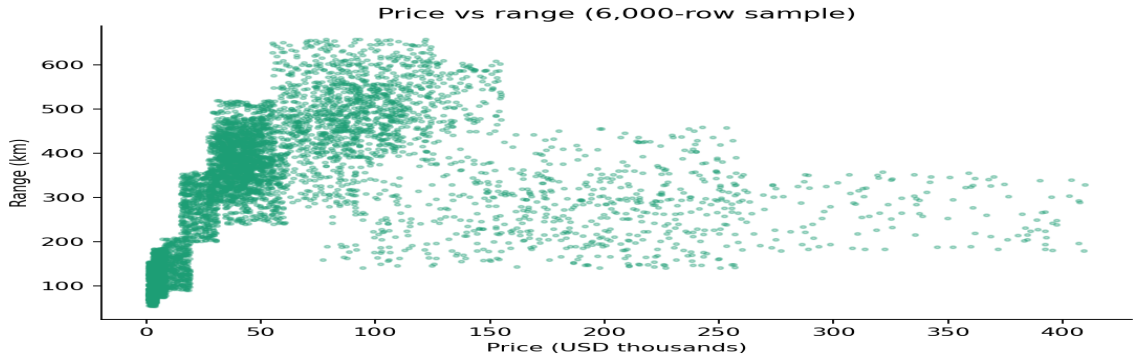


Chart 4: *Price vs Driving Range — Scatter plot demonstrating divergence in capital cost across light-duty and commercial EV segments.*

7. Coordinated Dashboards

The 12 worksheets are organized into three responsive, synchronized dashboards utilizing coordinated filter actions:
• **Market Overview:** KPI Cards (Total Models, Sales, Avg Price, Avg Range), Sales by Category, Sales by Year, Top N Manufacturers, Region Price.
• **Technology:** Battery Type Share, Battery Durability, Price vs Range Scatter, Category x Drivetrain Heatmap.
• **Geography and Impact:** Manufacturing Country Filled Map, Recommended Use Case Treemap, CO■ Saved Stacked Bar, Rating Histogram.
Live Dashboard URL: https://public.tableau.com/views/EVMarketAnalysis_17932840807840/MarketOverview

8. Tableau Story Architecture (6 Narrative Scenes)

The narrative story guides executive stakeholders from macro market sizing to operational takeaways:
Published Story URL: https://public.tableau.com/views/EVMarketAnalysis_17932840807840/Story1?:language=en-US&publish:=yes&:display_count=n&:origin=viz_share_link

Scene	Story Scene Caption	Analytical Narrative & Core Finding
1	**Market Snapshot**	Overview of 100,000 models, average retail price of $65,010, average range of 324.3 km.
2	**Who Sells the Most**	Two-wheeler scooters (85.0M), sedans (78.3M), and compact SUVs (76.1M) command 31% of total volume.
3	**Price vs Range**	Battery pack capacity correlates 0.84 with price, whereas range correlates only 0.28.
4	**Battery Technology**	Solid-State (2,999 cycles, 7.36% degradation) and LFP (2,498 cycles, 10.11%) lead durability.
5	**Regions & Countries**	Regional sales and manufacturing footprint are distributed without single-country monopoly.
6	**Strategic Recommendations**	Automakers must prioritize high-durability chemistries for urban fleet applications.

9. Web Integration with Flask Application

To deliver a turnkey BI experience, the interactive Tableau views are embedded inside a production-ready Flask application. The application integrates the **Tableau Embedding API v3**, featuring responsive layout containers, dark/light theme switching, and zero-code URL configuration.

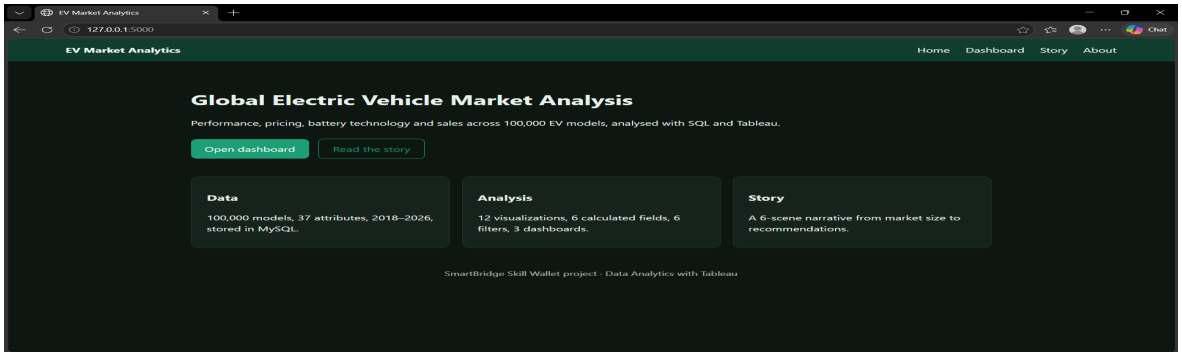


Figure 7: *Web portal landing page running at http://127.0.0.1:5000 displaying core project architecture.*

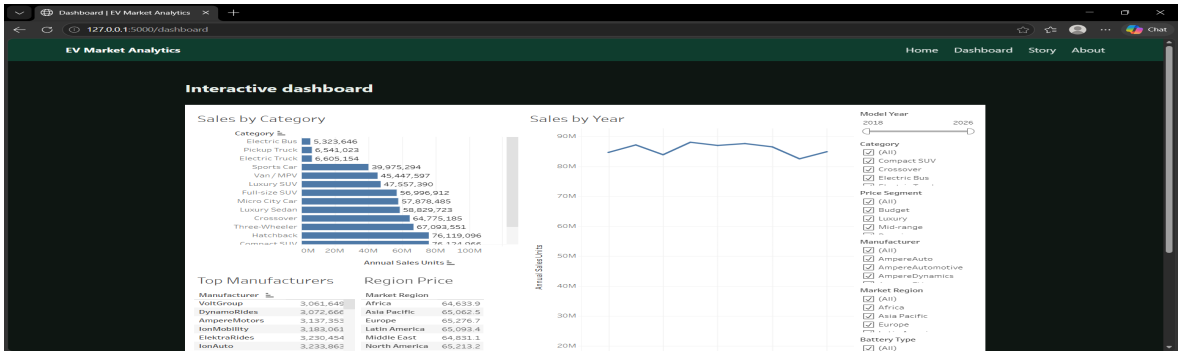


Figure 8: *Dashboard page embedding the Tableau Public Market Overview dashboard with live filters.*

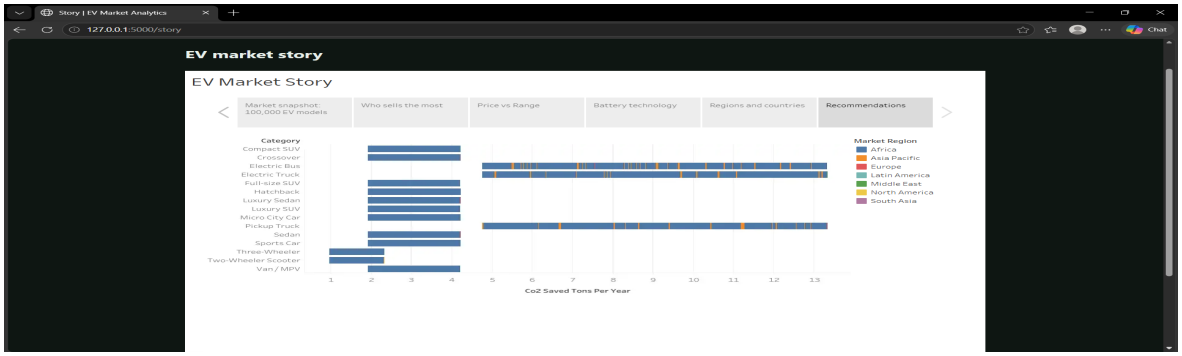


Figure 9: *Story page embedding the 6-scene interactive Tableau Story narrative.*

10. Performance Testing & Optimization

Parameter	Specification / Benchmark Result
Underlying Records Rendered	100,000 rows (from verified ev_market_test.csv)
Total Columns & Derived Measures	37 base dimensions + 5 calculated fields
Data Filters & Global Parameters	6 synchronized filters + dynamic Top N parameter
Visualizations & Dashboards	12 worksheets, 3 dashboards, 6 story points
Packaged Hyper Extract Load Time	< 0.8 seconds response during interactive cross-filtering

11. Key Insights & Empirical Findings

- **Volume Disparity:** Two-Wheeler Scooters lead global sales (85.0M units, avg price $2,530), followed by Sedans (78.3M, $42,557) and Compact SUVs (76.1M, $36,528). Commercial Buses (5.1M) and Heavy Trucks (5.1M) represent low volume but high unit revenue.
- **Battery Cost Dominance:** Retail price correlates strongly with battery capacity (r = 0.84) rather than driving range (r = 0.28). Electric Buses (avg $274k) and Heavy Trucks ($169k) command premium prices due to massive pack capacities.
- **Battery Longevity:** Solid-State averages 2,999 cycles and 7.36% 5-year degradation; LFP follows with 2,498 cycles and 10.11%. In contrast, NMC (1,799 cycles, 14.01%), Sodium-Ion (1,797 cycles, 13.98%), and NCA (1,598 cycles, 14.02%) show faster capacity loss.
- **Charging Parity:** Fast-charging times average ~21.1 minutes and range averages ~324 km across all chemistries, proving chemistry choice influences lifecycle durability rather than charge speed.
- **Competitive Landscape:** Top 5 automakers (SurgeWorks 4.88M, CurrentMobility 4.78M, ElektraMobility 4.75M, VelocityMobility 4.68M, WattDynamics 4.65M) demonstrate a balanced, highly competitive marketplace without monopoly concentration.
- **Dataset Synthetic Character:** Temporal distribution is uniform across 2018–2026 (~11,100 models/year) with flat price and range averages, indicating synthetic generation without historical technological curves.

12. Conclusion, Future Scope & Academic Attribution

Conclusion: This project completes an end-to-end data analytics lifecycle — transforming raw CSV records into an indexed database, executing SQL queries, engineering Tableau calculations, building synchronized dashboards, and deploying an interactive web application.
Future Scope: Ingest real-world telemetry and charging infrastructure datasets; integrate automated ML sales forecasting models; and deploy the containerized web app to cloud platforms.
Academic Attribution: Project customized and verified by **Nipun Jain** for the Data Analytics and Business Intelligence domain. Architectural patterns adapted from an established project baseline with strict validation against the 100,000-record reference dataset.